

```
In [2]: !pip install fg-data-profiling pandas numpy --quiet
```

```
In [3]: import pandas as pd
import numpy as np
import data_profiling
from data_profiling import ProfileReport
```

```
In [54]: #Loading data set to be reviewed and cleaned
df = pd.read_csv('My_Projects/Urban_Forestry/D365/311_SRs_2020.csv')
```

```
In [55]: #Generating an initial Profile Report of the data set to obtain a baseline understanding of the structure
profile = ProfileReport(df, title="Data Profile Report", explorative=True)
profile.to_notebook_iframe()
```

Summarize dataset: 0%| | 0/5 [00:00<?, ?it/s]

0%	0/13 [00:00<?, ?it/s]
8%	1/13 [00:00<00:02, 4.37it/s]
15%	2/13 [00:00<00:01, 5.84it/s]
23%	3/13 [00:01<00:04, 2.32it/s]
62%	8/13 [00:01<00:00, 6.44it/s]
69%	9/13 [00:04<00:02, 1.58it/s]
100%	13/13 [00:04<00:00, 2.85it/s]

Generate report structure: 0%| | 0/1 [00:00<?, ?it/s]

Render HTML: 0%| | 0/1 [00:00<?, ?it/s]

Data Profile Report



Overview

[Overview](#)[Alerts](#) 9[Reproduction](#)

Dataset statistics

Number of variables	13
Number of observations	5135
Missing cells	5146
Missing cells (%)	7.7%
Duplicate rows	0
Duplicate rows (%)	0.0%
Total size in memory	4.3 MiB
Average record size in memory	869.0 B

Variable types

Text	4
DateTime	3
Unsupported	1
Categorical	5

Variables

[Select Columns](#)

(Do Not Modify) Case

In [56]: *#Printing another structural overview of the data set*
`df.info()`

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 5135 entries, 0 to 5134
Data columns (total 13 columns):
 #   Column                                                                 Non-Null Count  Dtype
---  -
 0   (Do Not Modify) Case                                                5135 non-null   object
 1   (Do Not Modify) Row Checksum                                         5135 non-null   object
 2   (Do Not Modify) Modified On                                         5135 non-null   object
 3   Case Number                                                          5135 non-null   object
 4   Alternate Case Number                                                0 non-null      float64
 5   Service Request Type                                                 5135 non-null   object
 6   Department (Service Request Type) (Service Request Type)          5135 non-null   object
 7   Street Address                                                       5129 non-null   object
 8   Failure Time (Resolve SLA) (SLA KPI Instance)                     5130 non-null   object
 9   Status Reason                                                        5135 non-null   object
10   Status                                                              5135 non-null   object
11   Created On                                                          5135 non-null   object
12   Created By                                                          5135 non-null   object
dtypes: float64(1), object(12)
memory usage: 521.7+ KB
```

In [57]: *#Reviewing the first 5 rows of the data set*
`df.head()`

Out[57]:

	(Do Not Modify) Case	(Do Not Modify) Row Checksum	(Do Not Modify) Modified On	Case Number	Alternate Case Number	Service Request Type	Department (Service Request Type) (Service Request Type)
0	3239ff4c- aabe-461f- 80ad- 041395883cbf	bSfbWv9BQyVsF+PxIMeYkhWjUNHVoHashTrdp6l1p40PrD...	7/18/2023 2:21	101004466669	NaN	Tree Removal	Parks and Recreation
1	a4651bc2- 3734-4af1- bbeb- 34a2ccd30087	sr+d8ChblOR4P8hz4MNkk+GD9b/nCJqSML+f+UkOPktLc5...	7/22/2023 4:20	176970- 101004466592	NaN	Tree Trim	Parks and Recreation
2	caf37494- 2209-4634- 9fb5- 97c1692b44c7	cxNFSEnkOQwazFEvKJzxoJIMA5dHQzIPlxbU9PGDrc3E0Y...	7/18/2023 3:43	176969- 101004466564	NaN	Tree Removal	Parks and Recreation
3	0598d7d3- bc33-486f- a684- 7ef21defd926	G72AjYfd8hewXapSZA/8zXy4Ai8jFWlczjSiwhYmBNAwDz...	7/22/2023 18:04	176968- 101004466553	NaN	Tree Removal	Parks and Recreation
4	382e8e10- 4330-4e72- ab97- af3925736b6e	mQ9R7AwFWqzBj3SCzT6jN8nJLf/QyC4ULoguQl7IPDPR6U...	7/20/2023 20:31	176967- 101004466360	NaN	Tree Trim	Parks and Recreation

```
In [58]: #Calculating the sum of all null values for each column
df.isna().sum()
```

```

Out[58]: (Do Not Modify) Case 0
          (Do Not Modify) Row Checksum 0
          (Do Not Modify) Modified On 0
          Case Number 0
          Alternate Case Number 5135
          Service Request Type 0
          Department (Service Request Type) (Service Request Type) 0
          Street Address 6
          Failure Time (Resolve SLA) (SLA KPI Instance) 5
          Status Reason 0
          Status 0
          Created On 0
          Created By 0
          dtype: int64

```

```

In [59]: #The value for Case Number is the D365 case number combined with a prepended 6-digit Cityworks SR number
#When the two systems fail to interface, the prepended 6-digit Cityworks SR number is not combined
#Splitting the Case Number column into two new columns - SR_Num and Case_Num - to isolate the prepended
df2 = pd.read_csv('My_Projects/Urban_Forestry/D365/311_SRs_2020.csv')
df2[['SR_Num', 'Case_Num']] = df2['Case Number'].str.split('-', expand=True)
df2

```

Out[59]:

	(Do Not Modify) Case	(Do Not Modify) Row Checksum	(Do Not Modify) Modified On	Case Number	Alternate Case Number	Service Request Type	Depart (Se Re -
0	3239ff4c- aabe-461f- 80ad- 041395883cbf	bSfbWv9BQyVsF+PxIMeYkhWjUNHVoHashTrdp6l1p40PrD...	7/18/2023 2:21	101004466669	NaN	Tree Removal	Park Recre
1	a4651bc2- 3734-4af1- bbeb- 34a2ccd30087	sr+d8ChblOR4P8hz4MNkk+GD9b/nCJqSML+f+UkOPktLc5...	7/22/2023 4:20	176970- 101004466592	NaN	Tree Trim	Park Recre
2	caf37494- 2209-4634- 9fb5- 97c1692b44c7	cxNFSEnKOQwazFEvKJzxoJlMA5dHQzIPlxbU9PGDrc3E0Y...	7/18/2023 3:43	176969- 101004466564	NaN	Tree Removal	Park Recre
3	0598d7d3- bc33-486f- a684- 7ef21defd926	G72AjYfd8hewXapSZA/8zXy4Ai8jFWlcZjSiwhYmBNAwDz...	7/22/2023 18:04	176968- 101004466553	NaN	Tree Removal	Park Recre
4	382e8e10- 4330-4e72- ab97- af3925736b6e	mQ9R7AwFWqzBj3SCzT6jN8nJLf/QyC4ULoguQl7IPDPR6U...	7/20/2023 20:31	176967- 101004466360	NaN	Tree Trim	Park Recre
...	...	...	...	...	...	...	...
5130	f8f92f69- 7c93-472c- b2fa- 3de4a362cd58	9gzKS5MgBYJAEuo+ZJFD+TDOLy+FO1ajbV+5BNSxZ0jma4...	7/19/2023 18:34	169943- 101003786368	NaN	Tree Trim	Park Recre
5131	be680be3- e027-4b54- 819a- 9d15f43caba8	17iQjUzP6mvpRxqo5J22mfRLPm5upbvzA1o37DJjWbTtAp...	8/8/2021 5:40	169942- 101003786335	NaN	Tree Removal	Park Recre
5132	018233a2- aeb0-47d3-	fHwbBFa9OjtEm7RurbQ2DT0DzmEPaPTi0xG0TJaFP5koVf...	7/19/2023 22:17	169940- 101003786143	NaN	Tree Trim	Park Recre

	(Do Not Modify) Case	(Do Not Modify) Row Checksum	(Do Not Modify) Modified On	Case Number	Alternate Case Number	Service Request Type	Depart (Se Re ' (Se Re '
	b1d3- 2646d1cd53a6						
5133	ceb79339- b672-44f4- be20- 459ad2411aa1	i0dc7Kbkulf2aeuL7l/BO8MFGtA8CGOXn7RyUAnNOYf4zk...	7/19/2023 22:13	169938- 101003786039	NaN	Tree Removal	Park Recre
5134	306b642a- ee69-423a- 9387- 2bc0af856145	RUK4AAA7FmlzvFXEWJHtCyxO8TcsxX6cnZfGobHMydDXlc...	7/23/2023 0:11	169937- 101003785879	NaN	Tree Trim	Park Recre

5135 rows × 15 columns

```
In [60]: #Dropping unwanted columns
df3 = df2.drop(labels = ['(Do Not Modify) Case', '(Do Not Modify) Row Checksum', 'Alternate Case Number', '
df3
```

Out[60]:

	Case Number	Service Request Type	Department (Service Request Type) (Service Request Type)	Street Address	Failure Time (Resolve SLA) (SLA KPI Instance)	Status Reason	Status	Created On	SR_Num	Case_Num
0	101004466669	Tree Removal	Parks and Recreation	3400 BENFIELD DR&12934 WESTPARK DR	2/20/2021 19:57	Service Completed	Resolved	12/31/2020 19:56	101004466669	None
1	176970-101004466592	Tree Trim	Parks and Recreation	4540 W 34TH ST	3/16/2021 16:21	Service Completed	Resolved	12/31/2020 15:20	176970	101004466592
2	176969-101004466564	Tree Removal	Parks and Recreation	11830 KIRKHOLM DR	2/20/2021 14:08	Service Completed	Resolved	12/31/2020 14:07	176969	101004466564
3	176968-101004466553	Tree Removal	Parks and Recreation	9700 BECKLEY ST&1900 WAVELL ST	2/20/2021 13:43	Service Completed	Resolved	12/31/2020 13:42	176968	101004466553
4	176967-101004466360	Tree Trim	Parks and Recreation	12510 ASHLING DR	3/16/2021 10:31	Service Completed	Resolved	12/31/2020 9:30	176967	101004466360
...	...	...	...	...	...	...	...	...	...	...
5130	169943-101003786368	Tree Trim	Parks and Recreation	10119 GLENMAWR DR	3/17/2020 6:02	Service Completed	Resolved	1/2/2020 5:01	169943	101003786368
5131	169942-101003786335	Tree Removal	Parks and Recreation	425 HEIGHTS BLVD	2/22/2020 4:46	Service Completed	Resolved	1/2/2020 4:45	169942	101003786335
5132	169940-101003786143	Tree Trim	Parks and Recreation	1709 MARSHALL ST	3/17/2020 3:51	Service Completed	Resolved	1/2/2020 2:50	169940	101003786143
5133	169938-101003786039	Tree Removal	Parks and Recreation	5803 CHEENA DR	2/21/2020 18:10	Service Completed	Resolved	1/1/2020 18:09	169938	101003786039

	Case Number	Service Request Type	Department (Service Request Type) (Service Request Type)	Street Address	Failure Time (Resolve SLA) (SLA KPI Instance)	Status Reason	Status	Created On	SR_Num	Case_Num
5134	169937-101003785879	Tree Trim	Parks and Recreation	700 W DREW ST & 2200 STANFORD ST	3/16/2020 7:30	Service Completed	Resolved	1/1/2020 6:29	169937	101003785879

5135 rows × 10 columns

```
In [61]: #Filtering out all null values from Case_Num
df3['Case_Num'].isnull()
```

```
Out[61]: 0      True
1      False
2      False
3      False
4      False
...
5130   False
5131   False
5132   False
5133   False
5134   False
Name: Case_Num, Length: 5135, dtype: bool
```

```
In [62]: #Displaying the filter as a dataframe to determine the number of rows with null values
df3[df3['Case_Num'].isnull()]
```

Out[62]:

	Case Number	Service Request Type	Department (Service Request Type) (Service Request Type)	Street Address	Failure Time (Resolve SLA) (SLA KPI Instance)	Status Reason	Status	Created On	SR_Num	Case_Num
0	101004466669	Tree Removal	Parks and Recreation	3400 BENFIELD DR&12934 WESTPARK DR	2/20/2021 19:57	Service Completed	Resolved	12/31/2020 19:56	101004466669	None
15	101004465663	Tree Removal	Parks and Recreation	1827 RICHMOND AVE	2/20/2021 2:53	Service Completed	Resolved	12/31/2020 2:52	101004465663	None
23	101004464516	Tree Trim	Parks and Recreation	11531 FM 1960 RD	3/15/2021 4:39	Service Completed	Resolved	12/30/2020 3:38	101004464516	None
25	101004463973	Tree Removal	Parks and Recreation	12211 PINE KNOLL DR	2/18/2021 11:31	Service Completed	Resolved	12/29/2020 11:30	101004463973	None
26	101004463753	Tree Trim	Parks and Recreation	10006 AVES ST	3/14/2021 10:33	Service Completed	Resolved	12/29/2020 9:32	101004463753	None
...	...	...	...	...	...	...	...	...	...	...
4969	101003827009	Tree Code Violation	Parks and Recreation	2026 MACARTHUR ST	3/12/2020 17:34	Service Completed	Resolved	1/20/2020 16:33	101003827009	None
4998	101003822537	Tree Removal	Parks and Recreation	7116 ARBOR OAK DR&4300 CYPRESS GROVE LN	3/6/2020 9:37	Service Completed	Resolved	1/15/2020 9:36	101003822537	None
5001	101003822445	Tree Trim	Parks and Recreation	7315 HURTGEN FOREST RD	3/30/2020 9:38	Service Completed	Resolved	1/15/2020 8:37	101003822445	None
5007	101003822034	Tree Trim	Parks and Recreation		3/30/2020 5:47	Service Completed	Resolved	1/15/2020 4:46	101003822034	None
5110	101003789999	Tree Removal	Parks and Recreation	NaN	2/26/2020 3:47	Service Completed	Resolved	1/6/2020 3:46	101003789999	None

499 rows × 10 columns

```
In [63]: #Replacing all null values from Case_Num with values from SR_Num  
df3['Case_Num'] = df3['Case_Num'].fillna(df3['SR_Num'])  
df3
```

Out[63]:

	Case Number	Service Request Type	Department (Service Request Type) (Service Request Type)	Street Address	Failure Time (Resolve SLA) (SLA KPI Instance)	Status Reason	Status	Created On	SR_Num	Case_Num
0	101004466669	Tree Removal	Parks and Recreation	3400 BENFIELD DR&12934 WESTPARK DR	2/20/2021 19:57	Service Completed	Resolved	12/31/2020 19:56	101004466669	101004466669
1	176970-101004466592	Tree Trim	Parks and Recreation	4540 W 34TH ST	3/16/2021 16:21	Service Completed	Resolved	12/31/2020 15:20	176970	101004466592
2	176969-101004466564	Tree Removal	Parks and Recreation	11830 KIRKHOLM DR	2/20/2021 14:08	Service Completed	Resolved	12/31/2020 14:07	176969	101004466564
3	176968-101004466553	Tree Removal	Parks and Recreation	9700 BECKLEY ST&1900 WAVELL ST	2/20/2021 13:43	Service Completed	Resolved	12/31/2020 13:42	176968	101004466553
4	176967-101004466360	Tree Trim	Parks and Recreation	12510 ASHLING DR	3/16/2021 10:31	Service Completed	Resolved	12/31/2020 9:30	176967	101004466360
...	...	...	...	...	...	...	...	...	...	...
5130	169943-101003786368	Tree Trim	Parks and Recreation	10119 GLENMAWR DR	3/17/2020 6:02	Service Completed	Resolved	1/2/2020 5:01	169943	101003786368
5131	169942-101003786335	Tree Removal	Parks and Recreation	425 HEIGHTS BLVD	2/22/2020 4:46	Service Completed	Resolved	1/2/2020 4:45	169942	101003786335
5132	169940-101003786143	Tree Trim	Parks and Recreation	1709 MARSHALL ST	3/17/2020 3:51	Service Completed	Resolved	1/2/2020 2:50	169940	101003786143
5133	169938-101003786039	Tree Removal	Parks and Recreation	5803 CHEENA DR	2/21/2020 18:10	Service Completed	Resolved	1/1/2020 18:09	169938	101003786039

	Case Number	Service Request Type	Department (Service Request Type) (Service Request Type)	Street Address	Failure Time (Resolve SLA) (SLA KPI Instance)	Status Reason	Status	Created On	SR_Num	Case_Num
5134	169937-101003785879	Tree Trim	Parks and Recreation	700 W DREW ST & 2200 STANFORD ST	3/16/2020 7:30	Service Completed	Resolved	1/1/2020 6:29	169937	101003785879

5135 rows × 10 columns

```
In [64]: #Filtering out records where SR-Num contains only the case number
df3[df3['SR_Num'].str.contains('10100')]
```

Out[64]:

	Case Number	Service Request Type	Department (Service Request Type) (Service Request Type)	Street Address	Failure Time (Resolve SLA) (SLA KPI Instance)	Status Reason	Status	Created On	SR_Num	Case_Nu
0	101004466669	Tree Removal	Parks and Recreation	3400 BENFIELD DR&12934 WESTPARK DR	2/20/2021 19:57	Service Completed	Resolved	12/31/2020 19:56	101004466669	10100446666
15	101004465663	Tree Removal	Parks and Recreation	1827 RICHMOND AVE	2/20/2021 2:53	Service Completed	Resolved	12/31/2020 2:52	101004465663	10100446566
23	101004464516	Tree Trim	Parks and Recreation	11531 FM 1960 RD	3/15/2021 4:39	Service Completed	Resolved	12/30/2020 3:38	101004464516	10100446451
25	101004463973	Tree Removal	Parks and Recreation	12211 PINE KNOLL DR	2/18/2021 11:31	Service Completed	Resolved	12/29/2020 11:30	101004463973	10100446397
26	101004463753	Tree Trim	Parks and Recreation	10006 AVES ST	3/14/2021 10:33	Service Completed	Resolved	12/29/2020 9:32	101004463753	10100446375
...	...	...	...	...	...	...	...	...	...	...
4969	101003827009	Tree Code Violation	Parks and Recreation	2026 MACARTHUR ST	3/12/2020 17:34	Service Completed	Resolved	1/20/2020 16:33	101003827009	10100382700
4998	101003822537	Tree Removal	Parks and Recreation	7116 ARBOR OAK DR&4300 CYPRESS GROVE LN	3/6/2020 9:37	Service Completed	Resolved	1/15/2020 9:36	101003822537	10100382253
5001	101003822445	Tree Trim	Parks and Recreation	7315 HURTGEN FOREST RD	3/30/2020 9:38	Service Completed	Resolved	1/15/2020 8:37	101003822445	10100382244
5007	101003822034	Tree Trim	Parks and Recreation		3/30/2020 5:47	Service Completed	Resolved	1/15/2020 4:46	101003822034	10100382203
5110	101003789999	Tree Removal	Parks and Recreation	NaN	2/26/2020 3:47	Service Completed	Resolved	1/6/2020 3:46	101003789999	10100378999

499 rows × 10 columns

```
In [65]: #Replacing SR_Num with blank values where cases failed to interface  
df3.loc[df3['SR_Num'].str.contains('10100', na=False), 'SR_Num'] = ''  
df3
```

Out[65]:

	Case Number	Service Request Type	Department (Service Request Type) (Service Request Type)	Street Address	Failure Time (Resolve SLA) (SLA KPI Instance)	Status Reason	Status	Created On	SR_Num	Case_Num
0	101004466669	Tree Removal	Parks and Recreation	3400 BENFIELD DR&12934 WESTPARK DR	2/20/2021 19:57	Service Completed	Resolved	12/31/2020 19:56		101004466669
1	176970-101004466592	Tree Trim	Parks and Recreation	4540 W 34TH ST	3/16/2021 16:21	Service Completed	Resolved	12/31/2020 15:20	176970	101004466592
2	176969-101004466564	Tree Removal	Parks and Recreation	11830 KIRKHOLM DR	2/20/2021 14:08	Service Completed	Resolved	12/31/2020 14:07	176969	101004466564
3	176968-101004466553	Tree Removal	Parks and Recreation	9700 BECKLEY ST&1900 WAVELL ST	2/20/2021 13:43	Service Completed	Resolved	12/31/2020 13:42	176968	101004466553
4	176967-101004466360	Tree Trim	Parks and Recreation	12510 ASHLING DR	3/16/2021 10:31	Service Completed	Resolved	12/31/2020 9:30	176967	101004466360
...	...	...	...	...	...	...	...	...	...	...
5130	169943-101003786368	Tree Trim	Parks and Recreation	10119 GLENMAWR DR	3/17/2020 6:02	Service Completed	Resolved	1/2/2020 5:01	169943	101003786368
5131	169942-101003786335	Tree Removal	Parks and Recreation	425 HEIGHTS BLVD	2/22/2020 4:46	Service Completed	Resolved	1/2/2020 4:45	169942	101003786335
5132	169940-101003786143	Tree Trim	Parks and Recreation	1709 MARSHALL ST	3/17/2020 3:51	Service Completed	Resolved	1/2/2020 2:50	169940	101003786143
5133	169938-101003786039	Tree Removal	Parks and Recreation	5803 CHEENA DR	2/21/2020 18:10	Service Completed	Resolved	1/1/2020 18:09	169938	101003786039

	Case Number	Service Request Type	Department (Service Request Type) (Service Request Type)	Street Address	Failure Time (Resolve SLA) (SLA KPI Instance)	Status Reason	Status	Created On	SR_Num	Case_Num
5134	169937-101003785879	Tree Trim	Parks and Recreation	700 W DREW ST & 2200 STANFORD ST	3/16/2020 7:30	Service Completed	Resolved	1/1/2020 6:29	169937	101003785879

5135 rows × 10 columns

```
In [66]: #Splitting 'Created On' field into two fields to separate date and time
df3[['Create_Date', 'Create_Time']] = df3['Created On'].str.split(' ', expand=True)
df3
```

Out[66]:

	Case Number	Service Request Type	Department (Service Request Type) (Service Request Type)	Street Address	Failure Time (Resolve SLA) (SLA KPI Instance)	Status Reason	Status	Created On	SR_Num	Case_Num	Cre
0	101004466669	Tree Removal	Parks and Recreation	3400 BENFIELD DR&12934 WESTPARK DR	2/20/2021 19:57	Service Completed	Resolved	12/31/2020 19:56		101004466669	12
1	176970-101004466592	Tree Trim	Parks and Recreation	4540 W 34TH ST	3/16/2021 16:21	Service Completed	Resolved	12/31/2020 15:20	176970	101004466592	12
2	176969-101004466564	Tree Removal	Parks and Recreation	11830 KIRKHOLM DR	2/20/2021 14:08	Service Completed	Resolved	12/31/2020 14:07	176969	101004466564	12
3	176968-101004466553	Tree Removal	Parks and Recreation	9700 BECKLEY ST&1900 WAVELL ST	2/20/2021 13:43	Service Completed	Resolved	12/31/2020 13:42	176968	101004466553	12
4	176967-101004466360	Tree Trim	Parks and Recreation	12510 ASHLING DR	3/16/2021 10:31	Service Completed	Resolved	12/31/2020 9:30	176967	101004466360	12
...	...	...	...	...	...	...	...	...	...	...	...
5130	169943-101003786368	Tree Trim	Parks and Recreation	10119 GLENMAWR DR	3/17/2020 6:02	Service Completed	Resolved	1/2/2020 5:01	169943	101003786368	
5131	169942-101003786335	Tree Removal	Parks and Recreation	425 HEIGHTS BLVD	2/22/2020 4:46	Service Completed	Resolved	1/2/2020 4:45	169942	101003786335	
5132	169940-101003786143	Tree Trim	Parks and Recreation	1709 MARSHALL ST	3/17/2020 3:51	Service Completed	Resolved	1/2/2020 2:50	169940	101003786143	
5133	169938-101003786039	Tree Removal	Parks and Recreation	5803 CHEENA DR	2/21/2020 18:10	Service Completed	Resolved	1/1/2020 18:09	169938	101003786039	

	Case Number	Service Request Type	Department (Service Request Type) (Service Request Type)	Street Address	Failure Time (Resolve SLA) (SLA KPI Instance)	Status Reason	Status	Created On	SR_Num	Case_Num	Cre
5134	169937-101003785879	Tree Trim	Parks and Recreation	700 W DREW ST & 2200 STANFORD ST	3/16/2020 7:30	Service Completed	Resolved	1/1/2020 6:29	169937	101003785879	

5135 rows × 12 columns

```
In [67]: #Splitting 'Created On' field into two fields to separate date and time
df3[['Failure_Date', 'Failure_Time']] = df3['Failure Time (Resolve SLA) (SLA KPI Instance)'].str.split('
df3
```

Out[67]:

	Case Number	Service Request Type	Department (Service Request Type) (Service Request Type)	Street Address	Failure Time (Resolve SLA) (SLA KPI Instance)	Status Reason	Status	Created On	SR_Num	Case_Num	Cre
0	101004466669	Tree Removal	Parks and Recreation	3400 BENFIELD DR&12934 WESTPARK DR	2/20/2021 19:57	Service Completed	Resolved	12/31/2020 19:56		101004466669	12
1	176970-101004466592	Tree Trim	Parks and Recreation	4540 W 34TH ST	3/16/2021 16:21	Service Completed	Resolved	12/31/2020 15:20	176970	101004466592	12
2	176969-101004466564	Tree Removal	Parks and Recreation	11830 KIRKHOLM DR	2/20/2021 14:08	Service Completed	Resolved	12/31/2020 14:07	176969	101004466564	12
3	176968-101004466553	Tree Removal	Parks and Recreation	9700 BECKLEY ST&1900 WAVELL ST	2/20/2021 13:43	Service Completed	Resolved	12/31/2020 13:42	176968	101004466553	12
4	176967-101004466360	Tree Trim	Parks and Recreation	12510 ASHLING DR	3/16/2021 10:31	Service Completed	Resolved	12/31/2020 9:30	176967	101004466360	12
...	...	...	...	...	...	...	...	...	...	...	...
5130	169943-101003786368	Tree Trim	Parks and Recreation	10119 GLENMAWR DR	3/17/2020 6:02	Service Completed	Resolved	1/2/2020 5:01	169943	101003786368	
5131	169942-101003786335	Tree Removal	Parks and Recreation	425 HEIGHTS BLVD	2/22/2020 4:46	Service Completed	Resolved	1/2/2020 4:45	169942	101003786335	
5132	169940-101003786143	Tree Trim	Parks and Recreation	1709 MARSHALL ST	3/17/2020 3:51	Service Completed	Resolved	1/2/2020 2:50	169940	101003786143	
5133	169938-101003786039	Tree Removal	Parks and Recreation	5803 CHEENA DR	2/21/2020 18:10	Service Completed	Resolved	1/1/2020 18:09	169938	101003786039	

	Case Number	Service Request Type	Department (Service Request Type) (Service Request Type)	Street Address	Failure Time (Resolve SLA) (SLA KPI Instance)	Status Reason	Status	Created On	SR_Num	Case_Num	Cre
5134	169937-101003785879	Tree Trim	Parks and Recreation	700 W DREW ST & 2200 STANFORD ST	3/16/2020 7:30	Service Completed	Resolved	1/1/2020 6:29	169937	101003785879	

5135 rows × 14 columns

```
In [68]: #Verifying data types for all fields
df3.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 5135 entries, 0 to 5134
Data columns (total 14 columns):
#   Column                                                                 Non-Null Count  Dtype
---  -
0   Case Number                                                            5135 non-null   object
1   Service Request Type                                                  5135 non-null   object
2   Department (Service Request Type) (Service Request Type)           5135 non-null   object
3   Street Address                                                         5129 non-null   object
4   Failure Time (Resolve SLA) (SLA KPI Instance)                       5130 non-null   object
5   Status Reason                                                         5135 non-null   object
6   Status                                                                5135 non-null   object
7   Created On                                                            5135 non-null   object
8   SR_Num                                                                5135 non-null   object
9   Case_Num                                                             5135 non-null   object
10  Create_Date                                                            5135 non-null   object
11  Create_Time                                                            5135 non-null   object
12  Failure_Date                                                           5130 non-null   object
13  Failure_Time                                                           5130 non-null   object
dtypes: object(14)
memory usage: 561.8+ KB
```

```
In [69]: #Converting 'Created On' field to datetime data type  
df3['Created On'] = pd.to_datetime(df3['Created On'],dayfirst=False)  
df3
```

Out[69]:

	Case Number	Service Request Type	Department (Service Request Type) (Service Request Type)	Street Address	Failure Time (Resolve SLA) (SLA KPI Instance)	Status Reason	Status	Created On	SR_Num	Case_Num	Create
0	101004466669	Tree Removal	Parks and Recreation	3400 BENFIELD DR&12934 WESTPARK DR	2/20/2021 19:57	Service Completed	Resolved	2020-12-31 19:56:00		101004466669	12/31
1	176970-101004466592	Tree Trim	Parks and Recreation	4540 W 34TH ST	3/16/2021 16:21	Service Completed	Resolved	2020-12-31 15:20:00	176970	101004466592	12/31
2	176969-101004466564	Tree Removal	Parks and Recreation	11830 KIRKHOLM DR	2/20/2021 14:08	Service Completed	Resolved	2020-12-31 14:07:00	176969	101004466564	12/31
3	176968-101004466553	Tree Removal	Parks and Recreation	9700 BECKLEY ST&1900 WAVELL ST	2/20/2021 13:43	Service Completed	Resolved	2020-12-31 13:42:00	176968	101004466553	12/31
4	176967-101004466360	Tree Trim	Parks and Recreation	12510 ASHLING DR	3/16/2021 10:31	Service Completed	Resolved	2020-12-31 09:30:00	176967	101004466360	12/31
...	...	...	...	...	...	...	...	...	...	...	...
5130	169943-101003786368	Tree Trim	Parks and Recreation	10119 GLENMAWR DR	3/17/2020 6:02	Service Completed	Resolved	2020-01-02 05:01:00	169943	101003786368	1/2
5131	169942-101003786335	Tree Removal	Parks and Recreation	425 HEIGHTS BLVD	2/22/2020 4:46	Service Completed	Resolved	2020-01-02 04:45:00	169942	101003786335	1/2
5132	169940-101003786143	Tree Trim	Parks and Recreation	1709 MARSHALL ST	3/17/2020 3:51	Service Completed	Resolved	2020-01-02 02:50:00	169940	101003786143	1/2
5133	169938-101003786039	Tree Removal	Parks and Recreation	5803 CHEENA DR	2/21/2020 18:10	Service Completed	Resolved	2020-01-01 18:09:00	169938	101003786039	1/1

	Case Number	Service Request Type	Department (Service Request Type) (Service Request Type)	Street Address	Failure Time (Resolve SLA) (SLA KPI Instance)	Status Reason	Status	Created On	SR_Num	Case_Num	Create
5134	169937-101003785879	Tree Trim	Parks and Recreation	700 W DREW ST & 2200 STANFORD ST	3/16/2020 7:30	Service Completed	Resolved	2020-01-01 06:29:00	169937	101003785879	1/1

5135 rows × 14 columns

```
In [70]: #Converting 'Create_Date' field to datetime data type
df3['Create_Date'] = pd.to_datetime(df3['Create_Date'], dayfirst=False)
df3
```

Out[70]:

	Case Number	Service Request Type	Department (Service Request Type) (Service Request Type)	Street Address	Failure Time (Resolve SLA) (SLA KPI Instance)	Status Reason	Status	Created On	SR_Num	Case_Num	Create
0	101004466669	Tree Removal	Parks and Recreation	3400 BENFIELD DR&12934 WESTPARK DR	2/20/2021 19:57	Service Completed	Resolved	2020-12-31 19:56:00		101004466669	2020-
1	176970-101004466592	Tree Trim	Parks and Recreation	4540 W 34TH ST	3/16/2021 16:21	Service Completed	Resolved	2020-12-31 15:20:00	176970	101004466592	2020-
2	176969-101004466564	Tree Removal	Parks and Recreation	11830 KIRKHOLM DR	2/20/2021 14:08	Service Completed	Resolved	2020-12-31 14:07:00	176969	101004466564	2020-
3	176968-101004466553	Tree Removal	Parks and Recreation	9700 BECKLEY ST&1900 WAVELL ST	2/20/2021 13:43	Service Completed	Resolved	2020-12-31 13:42:00	176968	101004466553	2020-
4	176967-101004466360	Tree Trim	Parks and Recreation	12510 ASHLING DR	3/16/2021 10:31	Service Completed	Resolved	2020-12-31 09:30:00	176967	101004466360	2020-
...	...	...	...	...	...	...	...	...	...	...	...
5130	169943-101003786368	Tree Trim	Parks and Recreation	10119 GLENMAWR DR	3/17/2020 6:02	Service Completed	Resolved	2020-01-02 05:01:00	169943	101003786368	2020-
5131	169942-101003786335	Tree Removal	Parks and Recreation	425 HEIGHTS BLVD	2/22/2020 4:46	Service Completed	Resolved	2020-01-02 04:45:00	169942	101003786335	2020-
5132	169940-101003786143	Tree Trim	Parks and Recreation	1709 MARSHALL ST	3/17/2020 3:51	Service Completed	Resolved	2020-01-02 02:50:00	169940	101003786143	2020-
5133	169938-101003786039	Tree Removal	Parks and Recreation	5803 CHEENA DR	2/21/2020 18:10	Service Completed	Resolved	2020-01-01 18:09:00	169938	101003786039	2020-

	Case Number	Service Request Type	Department (Service Request Type) (Service Request Type)	Street Address	Failure Time (Resolve SLA) (SLA KPI Instance)	Status Reason	Status	Created On	SR_Num	Case_Num	Create
5134	169937-101003785879	Tree Trim	Parks and Recreation	700 W DREW ST & 2200 STANFORD ST	3/16/2020 7:30	Service Completed	Resolved	2020-01-01 06:29:00	169937	101003785879	2020

5135 rows × 14 columns

```
In [71]: #Converting 'Failure_Date' field to datetime data type
df3['Failure_Date'] = pd.to_datetime(df3['Failure_Date'], dayfirst=False)
df3
```

Out[71]:

	Case Number	Service Request Type	Department (Service Request Type) (Service Request Type)	Street Address	Failure Time (Resolve SLA) (SLA KPI Instance)	Status Reason	Status	Created On	SR_Num	Case_Num	Create
0	101004466669	Tree Removal	Parks and Recreation	3400 BENFIELD DR&12934 WESTPARK DR	2/20/2021 19:57	Service Completed	Resolved	2020-12-31 19:56:00		101004466669	2020
1	176970-101004466592	Tree Trim	Parks and Recreation	4540 W 34TH ST	3/16/2021 16:21	Service Completed	Resolved	2020-12-31 15:20:00	176970	101004466592	2020
2	176969-101004466564	Tree Removal	Parks and Recreation	11830 KIRKHOLM DR	2/20/2021 14:08	Service Completed	Resolved	2020-12-31 14:07:00	176969	101004466564	2020
3	176968-101004466553	Tree Removal	Parks and Recreation	9700 BECKLEY ST&1900 WAVELL ST	2/20/2021 13:43	Service Completed	Resolved	2020-12-31 13:42:00	176968	101004466553	2020
4	176967-101004466360	Tree Trim	Parks and Recreation	12510 ASHLING DR	3/16/2021 10:31	Service Completed	Resolved	2020-12-31 09:30:00	176967	101004466360	2020
...	...	...	...	...	...	...	...	...	...	...	...
5130	169943-101003786368	Tree Trim	Parks and Recreation	10119 GLENMAWR DR	3/17/2020 6:02	Service Completed	Resolved	2020-01-02 05:01:00	169943	101003786368	2020
5131	169942-101003786335	Tree Removal	Parks and Recreation	425 HEIGHTS BLVD	2/22/2020 4:46	Service Completed	Resolved	2020-01-02 04:45:00	169942	101003786335	2020
5132	169940-101003786143	Tree Trim	Parks and Recreation	1709 MARSHALL ST	3/17/2020 3:51	Service Completed	Resolved	2020-01-02 02:50:00	169940	101003786143	2020
5133	169938-101003786039	Tree Removal	Parks and Recreation	5803 CHEENA DR	2/21/2020 18:10	Service Completed	Resolved	2020-01-01 18:09:00	169938	101003786039	2020

	Case Number	Service Request Type	Department (Service Request Type) (Service Request Type)	Street Address	Failure Time (Resolve SLA) (SLA KPI Instance)	Status Reason	Status	Created On	SR_Num	Case_Num	Create
5134	169937-101003785879	Tree Trim	Parks and Recreation	700 W DREW ST & 2200 STANFORD ST	3/16/2020 7:30	Service Completed	Resolved	2020-01-01 06:29:00	169937	101003785879	2020-

5135 rows x 14 columns

```
In [72]: #Verifying data types for all fields
df3.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 5135 entries, 0 to 5134
Data columns (total 14 columns):
#   Column                                                                 Non-Null Count  Dtype
---  -
0   Case Number                                                            5135 non-null   object
1   Service Request Type                                                  5135 non-null   object
2   Department (Service Request Type) (Service Request Type)           5135 non-null   object
3   Street Address                                                         5129 non-null   object
4   Failure Time (Resolve SLA) (SLA KPI Instance)                       5130 non-null   object
5   Status Reason                                                         5135 non-null   object
6   Status                                                                5135 non-null   object
7   Created On                                                            5135 non-null   datetime64[ns]
8   SR_Num                                                                5135 non-null   object
9   Case_Num                                                             5135 non-null   object
10  Create_Date                                                           5135 non-null   datetime64[ns]
11  Create_Time                                                           5135 non-null   object
12  Failure_Date                                                         5130 non-null   datetime64[ns]
13  Failure_Time                                                         5130 non-null   object
dtypes: datetime64[ns](3), object(11)
memory usage: 561.8+ KB
```

```
In [73]: #Saving cleaned data set as new csv file  
df3.to_csv('My_Projects/Urban_Forestry/D365/311_SRs_2020_cleaned.csv', index=False)
```

```
In [ ]:
```